

KARHIV, Ukraine

WMO: 343000

Lat: 49.9281N

Lon: 36.2825E

Elev: 155

StdP: 99.48

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.5	-16.5	-23.0	0.5	-19.0	-20.0	0.6	-16.1	10.3	-5.5	9.4	-5.3	3.2	60	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	33.0	19.4	31.0	18.9	29.2	18.5	21.1	28.6	20.4	27.5	19.7	26.5	4.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.0	23.6	18.1	13.2	23.2	17.2	12.5	22.6	61.9	28.7	59.4	27.4	57.1	26.4	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.5	-22.3	35.3	3.8	2.3	-25.0	37.0	-27.2	38.3	-29.3	39.6	-32.0	41.3		
(5)			-22.6	22.5	3.7	0.7	-25.2	23.0	-27.4	23.4	-29.5	23.8	-32.2	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-4.9	-3.8	1.5	9.9	16.2	19.9	22.0	21.1	15.3	8.3	2.1	-2.8	(6)
(7)		DBStd	10.76	5.86	5.90	4.54	4.31	4.43	3.76	3.31	3.77	4.23	4.61	4.94	5.62	(7)
(8)		HDD10.0	1903	462	388	266	53	5	0	0	0	6	88	239	396	(8)
(9)		HDD18.3	3822	720	621	523	254	94	27	6	14	109	313	487	654	(9)
(10)		CDD10.0	1466	0	0	1	50	198	297	373	344	165	35	3	0	(10)
(11)		CDD18.3	342	0	0	0	1	29	73	120	101	18	1	0	0	(11)
(12)		CDH23.3	3208	0	0	0	14	274	628	1105	1014	167	6	0	0	(12)
(13)		CDH26.7	1115	0	0	0	2	68	186	410	409	39	1	0	0	(13)
(14)	Wind	WSAvg	3.7	4.1	4.2	4.3	3.9	3.5	3.4	3.2	3.1	3.3	3.4	3.9	4.2	(14)
(15)	Precipitation	PrecAvg	536	41	32	33	33	49	57	62	47	48	42	44	43	(15)
(16)		PrecMax	746	126	73	96	98	146	138	158	159	121	117	140	130	(16)
(17)		PrecMin	323	2	3	2	3	1	7	4	0	2	0	4	5	(17)
(18)		PrecStd	103	26	17	20	22	30	32	35	36	32	28	31	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	8.0	15.9	25.2	31.0	32.9	35.4	36.6	30.2	23.4	15.2	9.2	(19)
(20)			MCWB	5.7	6.2	8.9	13.8	17.5	19.5	20.0	19.4	16.4	14.5	12.1	8.2	(20)
(21)		2%	DB	4.6	5.9	12.8	21.9	28.5	30.5	32.9	33.3	27.4	20.1	12.9	7.2	(21)
(22)			MCWB	3.8	4.7	7.2	12.5	16.9	19.1	19.9	19.1	15.6	13.5	10.8	6.4	(22)
(23)		5%	DB	3.1	4.0	10.0	19.8	26.5	28.9	30.9	30.9	25.1	17.3	10.8	5.4	(23)
(24)			MCWB	2.5	2.8	5.9	11.6	16.3	18.6	19.6	18.7	15.4	12.5	9.0	4.8	(24)
(25)		10%	DB	1.9	2.8	7.7	17.4	24.2	27.1	28.9	28.8	22.8	15.0	8.4	3.8	(25)
(26)			MCWB	1.3	1.9	4.8	10.5	15.6	18.1	19.0	18.2	14.7	11.7	7.2	3.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	6.3	10.0	15.1	19.8	21.6	22.1	21.8	19.1	16.4	13.0	8.4	(27)
(28)			MCDWB	6.4	7.2	14.3	22.8	26.8	28.6	29.7	29.7	25.8	20.6	15.0	9.1	(28)
(29)		2%	WB	3.9	4.7	7.9	13.6	18.3	20.6	21.3	20.7	17.6	14.7	11.1	6.4	(29)
(30)			MCDWB	4.3	5.5	11.2	19.7	25.4	27.2	28.9	28.5	23.6	18.4	12.6	7.0	(30)
(31)		5%	WB	2.5	3.0	6.3	12.5	17.3	19.8	20.6	19.8	16.5	13.2	9.2	4.7	(31)
(32)			MCDWB	2.9	3.6	9.1	17.8	23.9	26.2	27.9	27.5	21.9	16.5	10.3	5.2	(32)
(33)		10%	WB	1.2	1.9	5.0	11.4	16.4	19.0	19.9	19.0	15.5	11.9	7.3	3.0	(33)
(34)			MCDWB	1.7	2.5	7.3	16.2	22.4	25.2	26.6	26.4	20.7	14.8	8.2	3.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.8	5.7	6.8	9.6	10.5	10.0	10.2	10.9	10.0	8.0	5.1	4.5	(35)
(36)			MCDBR	3.9	5.5	11.0	13.1	13.1	12.0	12.7	13.9	13.6	11.7	7.6	4.8	(36)
(37)			MCWBR	3.7	4.8	6.7	6.3	5.3	4.7	4.0	4.3	5.6	6.5	5.6	4.5	(37)
(38)		5% WB	MCDBR	3.7	5.1	9.5	11.1	11.4	10.7	10.9	11.7	11.0	9.9	6.6	4.6	(38)
(39)			MCWBR	3.7	4.6	6.5	5.9	5.2	4.7	4.2	4.6	5.4	6.4	5.5	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.348	0.379	0.456	0.412	0.402	0.433	0.434	0.403	0.367	0.348	0.321	(40)	
(41)		taud	2.150	2.080	2.149	2.056	2.270	2.319	2.243	2.227	2.282	2.347	2.290	2.226	(41)	
(42)		Ebn at Noon	688	760	812	785	847	857	822	800	784	747	657	629	(42)	
(43)		Edh at Noon	80	110	124	152	128	123	131	127	108	85	71	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.94	1.87	3.06	4.26	5.73	6.12	5.99	5.31	3.61	2.03	0.88	0.59	(44)	
(45)		RadStd	0.19	0.34	0.47	0.42	0.45	0.50	0.40	0.41	0.44	0.33	0.18	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.84	N/A	N/A	N/A	N/A	N/A	-175	-236	+126	+57
(47) Regional (7 neighbors)		+0.76	N/A	N/A	+0.99	N/A	N/A	-158	-219	+114	+66

Nomenclature: See separate page